



**Verified Carbon
Standard**

CLEAN COOKSTOVE FOR GARO COMMUNITIES



GHE Impact Ventures Pvt Ltd

Document Prepared by Infinite Environmental Solutions LLP

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The proposed project aims at reducing the fuel wood consumption in traditional mud stoves by disseminating improved cook stoves (ICS) at a highly reduced cost of 200 INR, this cost is taken only for the purpose of rural development in the areas and creating a value to the product. The transition from traditional to improved cookstoves will help in achieving significant reduction firewood consumption that will enable climate change mitigation in a sustainable manner. Overall objective of the project is to improve the lives of the rural women by providing them with efficient cooking technology. This project will not only empower women, but also decrease fuelwood consumption at household level which will enable reduction of carbon emissions at a global scale. The project will ensure that just sources of renewable biomass are used that are in line with the requirements according to Glossary of CDM terms v 11.0.

The existing scenario prior to the implementation of the project activity involves the usage of traditional mud stoves with poor ventilation, causing excessive indoor air pollution (IAP) and has been a serious health risk for women and children who spend several hours in the kitchen. The State of Meghalaya has an abundance of forest reserves and the ethnic communities rely on firewood for cooking. The state has three hill regions of the united Khasi and Jaintia Hills and Garo Hills. Garo Hills comprises of five district and the communities rely on traditional cooking practices and sources firewood from nearby forests. Penetration of modern cooking mediums like LPG gas, electric induction etc. have remained out of bound for most households in the region. The project activity is aimed to implement 20,000 cookstoves in the Meghalaya state of India. The Project activity had started distribution of Improved Cookstoves from 19-October-2022. The estimated total emission reductions for this crediting period are 663,315 tCO₂e and average annual emission reduction is 94,759 tCO₂e/year.

Audit Type	Period	Program	VVB Name	Number of years
Validation/ Verification	(19-October-2022-/18-October -2029)	<u>VCS Program</u>	LGAI Technological Center, S.A. (Applus+)	7 Years
Verification	19-October-2022 - 30-April-2023	<u>VCS Program</u>	LGAI Technological Center, S.A. (Applus+)	0 Year 6 Months and 11 Days
Total	19-October-2022 to 30-April-2023	<u>VCS Program</u>	LGAI Technological Center, S.A. (Applus+)	7 years for validation 0 Year 6 Months and 11 Days for verification

Under this project total number of Improved Cookstove which would be distributed are 20,000 in which 5,000 cookstoves have been distributed from till June 2023, and the next lot of 5,000 are planned to be distributed till end of 2023, and rest 10,000 would be distributed in year 2024. All the 20,000 cookstove would be distributed within the project boundary i.e. Meghalaya state in North-East part of India.

S. No	Year	Number of ICS distributed	Area
1	Year 2022	1,828	Meghalaya
2	Year 2023 (till March)	3,172	Meghalaya
3	Year 2023 (from April to December)	5,000	Meghalaya
4	Year 2024	10,000	Meghalaya
Total		20,000	Meghalaya

The distribution of the further cookstoves would depend on the availability of funds and carbon revenue generated from the projects.

1.2 Sectoral Scope and Project Type

The project activity is not a grouped project. The project activity will have following parameters:

Sectoral scope 03 - Energy Demand

Project Type: Type II Energy Efficiency Improved Cookstove Project.

Methodology: VMR0006: Methodology for Installation of High-Efficiency Firewood Cookstoves.¹ Ver1.1

Project Proponent hereby confirms that the project is not a grouped project.

1.3 Project Eligibility

The project activity includes the distribution of improved cookstove units for households that will enable the houses to transition from traditional cooking method to a fast, cleaner and efficient cookstove. The cookstove is a single pot improved cooking device at the household level with the

¹ <https://verra.org/methodologies/vmr0006-methodology-for-installation-of-high-efficiency-firewood-cookstoves/>

minimum thermal efficiency of 38 % which replaces mud stove open cooking methods used in the baseline.

Non-renewable biomass has been used in the project region since 31 December 1989², and wood has been used as primary source for thermal energy in the area for cooking. The project activity is a voluntary initiative of the project implementer and meets all the requirements of VCS Standard 4.4.

Table 1 Project eligibility

No.	The Scope of VCS Program	Project eligibility
1.	The seven Kyoto Protocol greenhouse gases.	The project activity includes reduction of Carbon Dioxide emissions which is one of the seven, Kyoto Protocol greenhouse gases included under the scope of VCS program
2.	Ozone-depleting substances (ODS).	The project activity does not involve use of Ozone Depleting Substances
3.	Project activities supported by a methodology approved under the VCS Program through the methodology approval process	This project activity is supported by VMR0006-Methodology for Installation of High Efficiency Firewood Cookstoves, which is approved under the VCS Program through the methodology approval process.
4.	Project activities supported by a methodology approved under an approved GHG Program, unless explicitly excluded	As demonstrated above, this project activity is supported by a methodology approved under VCS Program.
5.	Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements.	The project activity does not involve in Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program

The Project Activity does not fall under the category of excluded projects in Table 1 of the VCS Standard v4.4 and is therefore eligible under the scope of the VCS Program:

Excluded Large Scale Activity in Non LDC	Applicability
Grid-connected electricity generation using hydro-power plants/units	N/A: The Project activity does not involve generation of electricity using hydro-power plants/units

² fsi.nic.in/documents/sfr_1987_hindi.pdf- page no.45

Grid-connected electricity generation using wind, geothermal, or solar power plants/units	N/A: The Project activity does not involve generation of electricity using wind, geothermal, or solar power plants/units
Utilization of recovered waste heat for, inter alia combined cycle electricity generation and the provision of heat for residential commercial or industrial use	N/A: The Project activity does not involve generation of heat for producing electricity.
Generation of electricity and/or thermal energy using biomass. This does not include efficiency improvements in thermal applications (e.g. cookstoves)	N/A: The Project activity does not involve generation of electricity and/or thermal energy using biomass
Generation of electricity and/or thermal energy using fossil fuels, including activities that involve switching from a higher carbon content fuel to a lower carbon content fuel	N/A: The Project activity does not involve generation of electricity and/or thermal energy using fossil fuels.
Replacement of electric lighting with more energy efficient electric lighting, such as the replacement of incandescent electrical bulbs with CFLs or LEDs	N/A: The Project activity do not include the replacement of electric lighting with more energy efficient lighting
Installation and/or replacement of electricity transmission lines and/or energy efficient transformers	N/A: The Project activity do not include the installation and/or replacement of electricity transmission lines and/or energy efficient transformers.
Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions	N/A: The Project activity reduces GHG emissions due to energy efficiency measures by introducing improved cookstoves

1.4 Project Design

Project Proponent hereby confirm that the project includes multiple locations or installation and not being developed as a grouped project

Eligibility Criteria

Not Applicable

1.5 Project Proponent

Organization name	GHE Impact Ventures Private Limited
Contact person	Mr. Jaideep Bansal
Title	Chief Executive Officer
Address	GHE Impact Ventures Private Limited
Telephone	+91-8146846677
Email	jaideep@ghe.co.in

1.6 Other Entities Involved in the Project

Organization name	Infinite Environmental Solutions LLP
Role in the project	Carbon Project Developer
Contact person	Mr. Jimmy Sah
Title	Chief Operating Officer
Address	214-2015, Milindas Manor, 2 RNT Marg, Indore
Telephone	+91-731-4050174
Email	jimmy@infisolutions.org

1.7 Ownership

The ownership of the project lies with GHE Impact Ventures Private Limited. Further the declaration form which was signed by end user (Primary Beneficiary: cookstove owner) to Project Proponent during the registration process includes the statutory clause: “the carbon credit generated from the usage of Improved Cookstove by the user would be transferred to GHE.”

The improved cookstoves distributed and installed under this project activity owned by GHE.

1.8 Project Start Date

The first clean cookstove was commissioned on 19-October-2022. Thus, the project starts date is 19-October-2022.

1.9 Project Crediting Period

Project Crediting Period: Total Number of Years 7 renewable crediting period. Being renewable twice after the end of first crediting period. Hence, total crediting period of project activity is 21 years

Start Date: 19-October-2022 (First operating day of the project activity clean cookstove).

End date: 18-October-2029

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

This project is a small-scale project as GHG emission reduction from the project is less than 300,000 ERs per year.

Project Scale	
Project	√
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 1	98,808
Year 2	97,436
Year 3	96,077
Year 4	94,732
Year 5	93,401
Year 6	92,083
Year 7	90,778
Total estimated ERs	663,315
Total number of crediting years	07 years
Average annual ERs	94,759

For the current monitoring period only 5000 cookstove has been distributed and estimated ERs for the same is 13,129.

1.11 Description of the Project Activity

The project will be broadly implemented in Meghalaya state of India, which is jurisdiction under host country of India. ICS distributed under this project activity will target Garo communities for rural population and households using traditional mud stoves and use open fires for cooking for a very long time.

The project activity will distribute ICS to households, a portable improved-efficiency Greenway Jumbo stove manufactured in India.

Each and every cookstove is provided a unique identification number/ serial number. This number will be recorded during the registration process together with the following information (as appropriate and as available):

- Name of ICS user or head of the household
- Address
- Phone number of ICS user or household, where available.
- Stove model
- Date of distribution/installation
- ICS serial number

The information collected will be stored in the electronic database which will serve as project database for project monitoring and sampling purposes. The improved cookstove distributed for this project activity will have an average lifespan of 7 years with a thermal efficiency of about 38%. The project will replace mud stoves used in the baseline with Improved and fuel-efficient improved cookstove. The participant Garo communities are poor rural households and are highly dependent on the forest to fulfil their resource demands. For fuelwood access the community is highly dependent on forests.

The project aims to reach marginal people with clean cooking solution access in using Improved Cookstoves. Project implementer trained people appropriately at the local level to cater for the households with stove maintenance and basic repairing. After the installation, representatives from the implementing entity perform paperwork and the stove is recorded in the database. A baseline survey was conducted and found that the people were using mud stove in the baseline for cooking.

Table 2 Technical Description of cookstove

S.No.	Parameters	Greenway Jumbo Stove
1	Size	12"x11.4"x11.6"
2	Loading Capacity	40 kg
3	Fuel Savings	65%
4	Smoke reduction	70%
5	Design	Ergonomic Front-Loading Design
6	Air Induction	Secondary Air Induction Mechanism

7	Material	Different types of Steel and Aluminium
8	Warranty	3 Years replacement warranty
9	Package Content	Greenway Jumbo Stove, Bakelite handles, Screws, Screw driver, Fuel Tray (Firewood Support)
10	Life Span	6 to 7 Years
11	Thermal Efficiency	38% (as per KIRDI, Kenya)
12	Fuel Type	Wood, Cow dung, Argo waste etc.
 Photograph of the Greenway jumbo cookstove		

The Improved cookstove model is fuel-efficient, resulting in a decrease in fuel use in comparison to conventional pre-project stoves, project activity also helps in reducing smoke and indoor emissions.

Design considerations for this project activity have also incorporated ergonomics and safety of the participants through manuals and training were provided to project participants for use of the ICS.

The project will promote an efficient cookstove which is an improved and efficient design. The main design improvement is the use of a pre-fabricated metallic combustion chamber. This combustion chamber ensures consistent quality and durability of the ICS cookstove and will improve the lifespan of the stove with consistent performance in terms of efficiency, reduction of indoor air pollution and emissions, and safety. The improved cookstove will burn the fuelwood more efficiently, this will not only reduce the baseline fuelwood consumption but would also reduce the smoke generated from the use of traditional cookstoves. The usage of improved cookstove would save time for the user and thus provide extra time for livelihood options. The use of improved cookstove will fulfil the similar thermal needs of user as per in the baseline scenario.

The project cookstove has a lifetime of 7 years which requires the stoves to be replaced after 7 years of installation.

1.12 Project Location

Project activity would be in Meghalaya state of India. Thus, the geographical area of project activity is considered as Meghalaya state of India.



Figure 1 Project Location: Meghalaya

Project aims to distribute improved cookstoves in Meghalaya,

The coordinates of the project location is

	Start	End
Latitude	25° 25' 22.86"N	25° 15' 11.44"N
Longitude	89° 57' 39.84"E	92° 41' 18.06"E

1.13 Conditions Prior to Project Initiation

The scenario existing prior to implementation of the project activity is the usage of traditional biomass in inefficient cookstoves with poor ventilation, causing excessive indoor air pollution (IAP) and has been a serious health risk for women and children who spend several hours in the kitchen. Further refer to section 3.4 Baseline Scenario for further information about the baseline scenario.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

There are no mandatory laws or regulations in the host country (India) requiring the introduction of cookstove in India, the project includes the distribution of cookstoves and is completely

voluntary. The ongoing project does not violate any laws, statutes or other regulatory frameworks in India.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project activity has not been registered and is not seeking registration under any other GHG emission program to avail carbon benefits during the crediting period of the project activity.

1.15.2 Projects Rejected by Other GHG Programs

The project proponent hereby corroborates that the project activity has not been rejected by any other GHG program.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project does not reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading.

1.16.2 Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related environmental credit, including renewable energy certificates from other carbon registries such as Clean Development Mechanism, Gold Standard for Global Goals and Global Carbon Council.

1.16.3 Supply Chain (Scope 3) Emissions

This section is not applicable

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

This project activity contributes to the following SDGs in India:

- SDG 3: Good Health and Well Being
- SDG 5: Gender Equality
- SDG 7: Affordable and Clean Energy
- SDG 8: Decent work and Economic Growth
- SDG 12: Responsible consumption and production
- SDG 13: Climate Action

- SDG 15: Life on Land

Project's contribution to Sustainable Development Goals-

The contributions of proposed project activity towards sustainable development are explained with indicators viz. social, economic, environmental, technological well-being, legislative and temporal as follows:

Environmental well-being

The project activity will result in the reduction of firewood consumption and emission of greenhouse gases and thus conserve forest and biodiversity.

Social well being

The project activity will pave the way for development and increase the social status living conditions and the prevailing living standard in the vicinity of the project activity thus resulting in empowering the nearby. Also, it will contribute to increase in local employment by employing skilled and un-skilled personnel for the operation and maintenance of the equipment.

The project will reduce the drudgery of women, timesaving and the use of saved time for other productive livelihood activities.

The amount of indoor pollutants from burning of biomass in the kitchen will be reduced. Less carbon-di-oxide, carbon mono oxide, black carbon will be emitted due to the decrease in total fuelwood consumption.

Economic well being

The project has created a business opportunity during the construction phase for local stakeholders such as suppliers, contractors etc. contributing to economic well-being aspects.

Further, the project also influences the creation of employment opportunities for local people, which would enhance their social status. Sufficiently enhance indoor air quality thereby improving the health of women and children and reducing incidences of smoke and fire-related injuries and therefore result in saving of health-related expenses.

Technological well being

The proposed project activity will promote improved cookstoves that result in reduced fuel consumption and emissions due to cooking and heating water in homes.

1.17.2 Sustainable Development Contributions Activity Monitoring

This project will provide 20,000 Clean cookstove to households in rural and hill tribe villages of Meghalaya for replacing the traditional mud stoves with open fire and contributes to Intended. Sustainable Development Contributions (SDGs) which are addressed by the project are as follows:

Table 3 Sustainable Development Goals addressed by the project

Sustainable Development Goals Targeted	SDG Impact	Estimated Annual Average
SDG 3: Good health and well being	Reduction in smoke related illness due to use of ICS instead of traditional cookstoves. Therefore, the SDG indicator 3.9.1 'mortality rate attributed to household and ambient air pollution' which will be accessed through 'improvement of air quality' in project households due to successful operation of Improved cookstove	20,000 households are targeted under the project activity. This will have a positive impact on the living conditions of the local communities particularly the marginalized.
SDG 5: Achieve gender equality and empower all women and girls	The project is empowering women to volunteer for community engagement and monitor the usage of the cookstove in the households. This activity meets the Target 5.5: "Ensure women's full and effective participation and equal opportunities for leadership at all levels of the decisions making in political, economic and public life	Till the period of first monitoring distribution of 5,000 cookstoves were done in which 35 women were involved in the distribution and monitoring of the project
SDG 7. Ensure access to affordable, reliable, sustainable and modern energy for all	The project is designed to provide cleaner, safer, more affordable and more efficient cooking equipment than traditional stoves to poor household by means of carbon finance. The indicator to monitor the access to affordable and clean energy is the increase in number of households using clean cooking devices such as Improved Cook Stoves. This project is aim to full fill the target 7.1: "Ensure access to affordable, reliable and modern energy services"	Usage of 20,000 improved cookstoves by the beneficiaries. Till the period of first monitoring 5,000 cookstoves have been distributed.
SDG 8: Decent Work and Economic Growth	The project will contribute towards below SGD target 8.5 & 8.5.1, "by 2030 achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.	Employment of local volunteers and local NGOs for management of the project
SDG 12. Ensure sustainable consumption and production patterns	The project will lead to reduction in fuelwood consumption on daily basis which was earlier directly harvested from the forest. The indicator to monitor the target is consumption of fuelwood by the project device The project is aiming to achieve target 12.2: "By 2030, achieve the sustainable management and efficient use of natural resources."	The improved cookstoves consume less fuelwood as compared to the mud stove used in baseline. fuel savings per-household in first year is 3.76 tonne/annum
SDG 13: Climate Action (mandatory)	Indicator taken by the project is 13.2.2 "Tonnes of greenhouse gas emissions avoided or removed."	Estimated 663,315 tCO ₂ in next seven years,
SDG 15. Sustainably manage forests, combat desertification, halt and reverse land degradation, halt biodiversity loss	The efficient cookstoves will use less fuelwood for the cooking task and thus reduce GHG emissions compared to the traditional way of cooking. The indicator to monitor the target is the amount of fuel wood consumed by the project device This project aims to target 15.2: "By 2020, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded	The improved cookstoves consume less fuelwood as compared to the mud stove used in baseline

	forests and substantially increase afforestation and reforestation globally"	
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Table 4: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
Sequential row number	SDG Target number	Number and text of SDG indicator or, if no official SDG indicator is applicable, user-defined indicator	Indicate the project's contribution to the SDG Indicator (implemented activities to increase or decrease)	Brief description of the quantifiable impact of the project's activities related to the SDG indicator, during the monitoring period.	Brief description of the cumulative quantifiable impact of the project's activities related to the SDG indicator, over the project lifetime.
1)	3.9	3.9.1 Mortality rate attributed to household and ambient air pollution	Implemented activities to decrease household air pollution	5,000 cookstoves have been distributed in this monitoring period.	5,000 cookstoves has been distributed in this monitoring period.
2)	5.5	5.5.2 Proportion of women in managerial position	Implemented activities to increase women participation	In this monitoring period around 35 women were trained for the distribution of the improved cookstoves	In this monitoring period around 35 women were trained for the distribution of the improved cookstoves
3)	7.1	7.1.2 Proportion of population with primary reliance on clean fuels and technology	Implemented activities to increase use of clean technology	In this monitoring period 5,000 improved cookstoves had been distributed amongst rural households.	In this monitoring period 5,000 improved cookstoves had been distributed amongst rural households.

4)	8.3	8.3.1 Proportion of informal employment in total employment, by sector and sex	Implemented activities to increase employment	35 people were employed in distribution of cookstove	35 people were employed in distribution of cookstove
5)	12.0	Tonne of fuelwood consumption for energy purposes	Implemented activities to decrease fuelwood consumption	1.725 tonne in each household is saved during the monitoring period	1.725 tonne in each household is saved during the monitoring period
6)	13.0	Tonne of greenhouse gas emissions avoided or removed	Implemented activities to decrease the green-house gas emissions	In this monitoring period, 11,376 tonne of CO ₂ e is reduced by the usage of improved cookstoves	In this monitoring period, 11,376 tonne of CO ₂ e is reduced by the usage of improved cookstoves
7)	15.2	15.2.1 Progress towards sustainable forest management	Implemented activities to decrease the fuelwood consumption	1.725 tonne in each household is saved during the monitoring period	1.725 tonne in each household is saved during the monitoring period

1.18 Additional Information Relevant to the Project

Leakage Management

For this project activity 0.95 as a discount factor is taken in to consideration for leakage management as per the requirement of the methodology VMR006 version 1.1.

Commercially Sensitive Information

Commercial sensitive information will be made available to VVB (Validation and Verification Board) on request.

Further Information

Not Applicable

2 SAFEGUARDS

2.1 No Net Harm

No negative impact of project activity. The project is implemented to reduce carbon emission from the open burning of fuelwood in mud stove. The project will only bring the positive impacts on environmental and social economic as described in section 1.17.

2.2 Local Stakeholder Consultation

The local stakeholder consultation under project activity was conducted on 8-November-2022 at Gasaupara village Gasaupara block of South Garo hills district. All interested stakeholders representing the following groups: traditional stove users, representatives from local self-help groups, field assistants working in the project, local leaders and NGO representatives attended the meeting.

The stakeholders were informed over phone and messaging applications app. Detailed contact information such as the phone number and email address of the representative of the project implementer has also made available for the stakeholders. A total of 89 participants attended the meeting. The attendance sheet for the same is described below:

SI No	Name	Gender	Designation	Village
1	Probinballa ch sangma	Female	VEC secretary	Meguasongmong
2	Nikche ch sangma	Female	VEC secretary	Chinengkare
3	Prodip marak	Male	Secretary	Mekagre
4	Allishachira marak	Female	Secretary	Rongsugre
5	Ejitean marak	Male	Secretary	Kangmejalegre
6	Sulikha sangma	Female	Secretary	Ruganokat
7	Julikha marak	Female	Joints account	Ruganokat
8	Subitha sangma	Female	Secretary	Bugadongre
9	Salnira momin	Female	Secretary	Kolapara
10	Sumari momin	Female	VEC President	Kolapara
11	Jasinth sangma	Female	Secretary	Kagasipara
12	Thishstand sangma	Male	Secretary	Mennangre
13	Tibith marak	Male	Secretary	Tebesugre
14	Mopochi D Sangma	Female	Village member	Ruatsongma
15	Modue sangma	Male	Secretary	Gandhibule
16	JSaac a sangma	Male	Secretary	Rongchonggre
17	Donalikka S Sangma	Female	VEC Secretary	Gobilakush
18	Isaya marak	Male	Secretary	Rongsengre
19	Tetush T Sangma	Male	Vec secretary	Chitillonggre

20	Ronjith sangma	Male	Secretary	Rongdukgre
21	Johnsheng Marak	Male	Vec secretary	Baigongau
22	Silbarth sangma	Male	Secretary	Tajangre
23	Nerula Marak	Female	President	Chengkurenggre
24	Wilbirth sangma	Male	BFT	Seenpara
25	Artrick sangma	Male	Secretary	Pikapara
26	Matseng Ch Marak	Male	President	Chengkurenggre
27	Nerula M Marak	Female	Secretary	Ruga songma
28	Philo sangma	Male	Secretary	Mamchapara
29	Chariers M Marak	Male	Secretary	Rangralantalu
30	Denobina sangma	Female	Secretary	Chankchigre
31	Silnang D Marak	Male	Secretary	Dusenggre
32	Sengme momin	Female	Secretary	Drongsongmong
33	Athina sangma	Female	Secretary	Kondok
34	Chamcheng Sangma	Male	Secretary	Konagiitim
35	Relshy marak	Female	Secretary	Meguabagre
36	Emlyan Momin	Male	Secretary	Mangkeng songma
37	Praibesh Sangma	Male	Secretary	Dollonggre
38	Silkame Ch Marak	Female	Secretary	Bamonpara
39	Namseng Sangma	Male	Secretary	Songkalak
40	Premika k Marak	Female	Secretary	Seempara
41	Lenish R Marak	Male	Secretary	Balnargre
42	Saltina sangma	Female	Secretary	Rugadomigre
43	Barmin M Sangma	Male	Secretary	Ramchengga
44	Jessica ch Sangma	Female	President	Namchapara
45	Paralad banai	Female	Secretary	Newjatnakam
46	Salbina ch Sangma	Female	Secretary	Omjora
47	Saltima sangma	Female	Secretary	Gandogre
48	Sibula Hajong	Female	Secretary	Debulapara
49	Raksera	Male	Secretary	Rugadomigre
50	Killingson marak	Male	Secretary	Mandargre
51	Benica Sangma	Female	Secretary	Darokgre
52	Namitha sangma	Female	Secretary	Aritakagre
53	Srangbirth Marak	Male	Village member	Rengguagre
54	Lebika sangma	Female	Secretary	Bandaprara
55	Romein P Sangma	Female	Village member	Dimagre
56	Ashapura banai	Female	Secretary	Jatnakora
57	Ro	Female	Robina T Sangma	Ganchibon
58	Henritha sangma	Female	Secretary	Cherenghungith
59	Protilla Marak	Female	Village member	Meninggre
60	Abenilash marak	Female	Village members	Dimargee
61	Salkimi sangma	Female	Secretary	Rongmasu
62	Sashulle R Marak	Female	Village member	Kondok
63	Anjona sangma	Female	Secretary	Jodugre
64	Eling marak	Male	Secretary	Rukepara

65	Binnne marak	Male	Secretary	Anglotsle
66	Terson marak	Male	Secretary	Rengguagre
67	Chelana marak	Female	Secretary	Dinagre
68	Silmera marak	Female	Secretary	Dinagre
69	Chilara sangma	Female	Village members	Gundhibor
70	Sharina D sangma	Female	Secretary	Dinachigitchak
71	Albenitha	Female	Secretary	Chengkali
72	Ritha marak	Female	Village members	Kondok
73	Ritha marak	Female	Village members	Kondok
74	Rosha sangma	Female	Secretary	Konkona
75	Lesheana ch sangma	Female	APO	Gasuapara
76	Medina D marak	Female	Account Asst	Gasuapara
77	Shibhom KR paul	Male	BPM	Gasuapara
78	Pronab marak	Male	GS	Dalu block
79	Peterkin sangma	Male	GS	Dalu block
80	Mark sangma	Male	GS	Dalu block
81	Aldrin marak	Male	GS	Dalu block
82	Fendinaned sangma	Male	GS	Gasuapara
83	Saph sangma	Male	GS	Gasuapara
84	Roerich marak	Male	GS	Gasuapara
85	Nomera momin	Female	VEC Secretary	Konchikolgre
86	Benpher sangma	Female	GS	Gasuapara
87	Prokanch hajong	Male	GS	Gasuapara
88	Robila marak	Female	VEC secretary	Kodimboka
89	Sanalishi Sangma	Male	VEC secretary	Makaliripara

The project representatives explained in detail about the improved cookstoves which are to be employed. The detailed aspects of the project activity including its objectives, associated benefits and environmental and socio-economic issues were also discussed.



Figure 2 Photograph from LSC



Figure 4 Photograph from LSC 2



Figure 3 Photograph form LSC 3

Before the implementation of this project, the stakeholder groups have been identified by Project Proponent, who knows the local households and customs best, which are listed as follows.

- 1) 20,000 target households These 20,000 households are directly affected by this project activity since they receive the clean cookstove to replace the traditional mud stove for cooking. They signed the End-user Agreement that agreed to transfer the legal right of emission reduction to GHE Impact Ventures Private Limited for free.
 - 2) GHE Impact Ventures Private Limited developed the project implementation, to organize the local stakeholders meeting and make the allocation plan of the greenway clean cookstove.
 - 3) GHE Impact Ventures Private Limited, At the start of this project, GHE Impact Ventures Private Limited are responsible to deliver the greenway clean cookstove to the 20,000 households and make the end-users' name list. Besides, during the operation period, they are required to support the monitoring activities, including connecting, organizing and guiding in questionnaire survey,
1. The process or method used for engaging local stakeholders (e.g., dates of announcement or meetings, periods during which input was sought). The stakeholder consultations include questionnaires and stakeholder meeting, requiring stakeholders to present their opinions on the implementation of project and their willingness to participate. Before the project started, the meeting invitation, including the meeting time and main topics, has been announced through

phone calls, emails and online apps to all the stakeholder groups. And then, the representatives of end-users,

In the stakeholders meeting on 08-November-2022, the representatives of govt. officials and the project proponent are responsible to explain further the potential costs, risks and benefits to relevant communities and stakeholders. After project proponent's detailed explanation, they know that the implementation of the project would generate GHG emission reductions, mitigate climate change, reduce smoke emissions, benefit children's education and bring additional job opportunities for local people by providing them with relevant technical skills and training. Thus, all the stakeholders agreed to conduct this project in the end.

2. The procedure or method used for documenting the outcomes of the local stakeholders The outcomes of the local stakeholders have been documented by two methods, which are minutes of stakeholder meeting and the questionnaires distributed in the meeting. Firstly, according to the minutes of stakeholder meeting, the agenda is as below.

- A. Opening of the meeting.
- B. Explanation of project (including the project design, applied technology, objectives, scale, contribution progress, operation plan, introduction to the GHE Impact Ventures Private Limited
- C. Presentation distribution processing of Clean Cookstove
- D. Sharing experience of distributors
- E. Question & Answers (Discussion and Feedback from participants)
- F. Demonstration of using, maintaining and giving the Greenway Clean Cookstove to local households.
- G. Conclude and agree on the minutes of the meeting Secondly, there are also questionnaires distributed to the attendees for collecting the comments from stakeholders.

3. The mechanism for on-going communication with local stakeholders the phone number of contact person appointed by project proponent has been noticed to the stakeholders in the meeting as well as email address for on-going communication with local stakeholders. Besides, if any comments raised, local stakeholders can communicate with the GHE Impact Ventures Private Limited.

Table 5 Feedback by stakeholders in LSC Round

Stakeholders' comment	Comments & Questions	Responses to the comments
Probinballa ch sangma VEC secretary Meguasongmong village	How many people in the family can be cooked by the greenway clean cookstove?	For 4-6 persons in a family.
Sumari momin VEC President Kolapara village	I realize that greenway clean cookstove is very convenient for the poor in our district, since most people cook with firewood. Can they be provided with more stoves?	The project has a plan to allocate greenway clean cookstove according to the necessary needs of each household in Garo hill area. The average person in household is 5 in Garo hill area, so each household is planned to be distributed one greenway clean cookstove.
Mopochi D Sangma Ruatsongma village	How much cooking time to increase in greenway clean cookstove?	Approx 40% of time would get reduced in cooking on greenway clean cookstove
Wilbirth sangma BFT Seenpara village	Please ensure that metal part do not overheat during cooking.	Metal part near the burner may overheat during cooking, but the handles on both sides of the greenway clean cookstove can be used to move it from one place to another to prevent people who cook from being scalded.

At the end of the meeting, the stakeholder understood that this project can provide higher thermal efficiency cookstoves, protect the environment, improve health and education, and provide additional job opportunities. Women particularly liked the idea of using small amounts of firewood, the stoves producing less smoke and saving them time. They are eager to start this project and see tangible benefits.

No comments or inputs were received during the consultation on the project design.

For Continuous input and grievance mechanism: The stakeholders were informed that the project has designed a continuous input and grievance mechanism to ensure all grievances are recorded and responded. The contact details and location for maintaining a log book to record the complaints of the stakeholders was also shared.

No negative inputs were received during the consultation; however, grievance procedure is in place and no grievance has been recorded during the monitoring period.

A grievance registers are also placed in the local ground offices too, stakeholders could connect with the field staffs and register any grievance for the project.

2.3 Environmental Impact

No negative environmental impacts have been identified from the project and environmental impact assessment (EIA) is not required for the project.

2.4 Public Comments

No Comments were raised during the public comment period. The public comment period was started 30-March-2023 to 29- April -2023.

2.5 AFOLU-Specific Safeguards

This section is not application as the project is a non-AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Following approved baseline & monitoring methodology is applied.

Methodology: VMR0006- Methodology for Installation of High Efficiency Firewood Cookstoves.
Version: 1.1. Sectoral Scope: 03

Tools: CDM TOOL30: Calculation of the fraction of non-renewable biomass”; Version-04.

Tools: CDM Tool 1: Tool for the demonstration and assessment of additionality Version 07.0.0

Tools: CDM Tool 21: Demonstration of additionality of small-scale project activities. Version 13.1

3.2 Applicability of Methodology

Applicability for VMR0006 version 1.1

S. No	VCS Methodology Requirement	Project Justification
Applicability for VMR0006		
01	Project activities shall be implemented in domestic premises, or in community-based kitchens.	The improved cookstove is distributed amongst the different household in the rural areas of Garo Hills in Meghalaya state of India.
02	The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer’s specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of project stove replacing fossil fuel baseline stove, it shall exclusively use renewable biomass.	In the project Jumbo cookstove by Greenway India is been distributed amongst the community. The thermal efficiency of the Greenway Jumbo cookstove is 38%- which is certified by third party
03	Both ‘Projects’ and ‘Large Projects’ can use this methodology	This is a small-scale project, hence the Methodology VMR0006 Ver 1.1 is applicable
04	Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to	Proof that biomass has been used in the project region, i.e., in the host country India since 31 December 1989 is available

	published literature, official reports or statistics	through widespread documentation. According to National Family Health Survey ³ in year 2019-2020, in India around 42% of total household still uses non-clean fuel for their daily cooking practise. And in rural area this number goes up to 57%.
05	For the specific case of biomass residues processed as a fuel (e.g., briquettes, wood chips), it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g., shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded	The clean cookstove is introduced as an energy efficiency measure to replace baseline stoves and reduce the use of non-renewable biomass for combustion. The improved cookstove consumes fuelwood. Hence, this is point is not applicable.
06	The project description shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g., programme logo).	A record-keeping system and unique identification of the improved cookstoves will be used. This will include username, user identification number and address/ location of the user's house and distribution date. The record-keeping system will ensure that each ICS can be traced to project instances to avoid double counting.
07	The project description shall also explain how the proposed procedures prevent double-counting of emission reductions, for example, to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	The project proponent will sign an end-user agreement with individual households/users to ensure only Project Proponent can claim the emission reduction. and Project Proponent will also sign the contract with the Manufacturer.

³http://rchiips.org/nfhs/NFHS-5_FCTS/Meghalaya.pdf

Applicability for Tool 30 version 04.0: Calculation of the fraction of non-renewable biomass

S. No	CDM Tool 30 Requirement	Project Justification
1.	DNAs to submit region- or country-specific default f_{NRB} values, following the procedures for development, revision, clarification and update of standardized baselines (SB procedures); or	The project location is Meghalaya. This tool is used to calculate the f_{NRB} values for the same. The f_{NRB} value would be used for determining the project emission reductions.

Applicability for Tool 01 version 07.0: tool for demonstration and assessment of additionality

S. No	CDM Tool 01 Requirement	Project Justification
1.	The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	Not applicable, as new methodology is not proposed in the project activity.
2	Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	The tool is not included in the methodology, but PP is using Simple cost analysis for establishing the additionality.

Applicability for Tool 21 version 13.1: Demonstration of additionality of small-scale project activities

S. No	CDM Tool 21 Requirement	Project Justification
1.	The use of the methodological tool “Demonstration of additionality of small-scale project activities” is not mandatory for project participants when proposing new methodologies.	Not applicable, as new methodology is not proposed in the project activity.

	Project participants and coordinating/managing entities may propose alternative methods to demonstrate additionality for consideration by the Executive Board.	
2	Project participants and coordinating/managing entities may also apply "TOOL19: Demonstration of additionality of microscale project activities" as applicable.	Since the project is not a micro-scale project hence, tool 19 is not applied.

3.3 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emission from use of non-renewable biomass/Fossil fuel	CO ₂	Yes	Major source
		CH ₄	Yes	Major source
		N ₂ O	Yes	Major source
		Other	No	No other Source Identified
Project	Emission from use of non-renewable biomass/Fossil fuel	CO ₂	Yes	Major source
		CH ₄	Yes	Major source
		N ₂ O	Yes	Major source
		Other	No	No other source identified

A representation of the baseline boundary and project boundary are given as below.

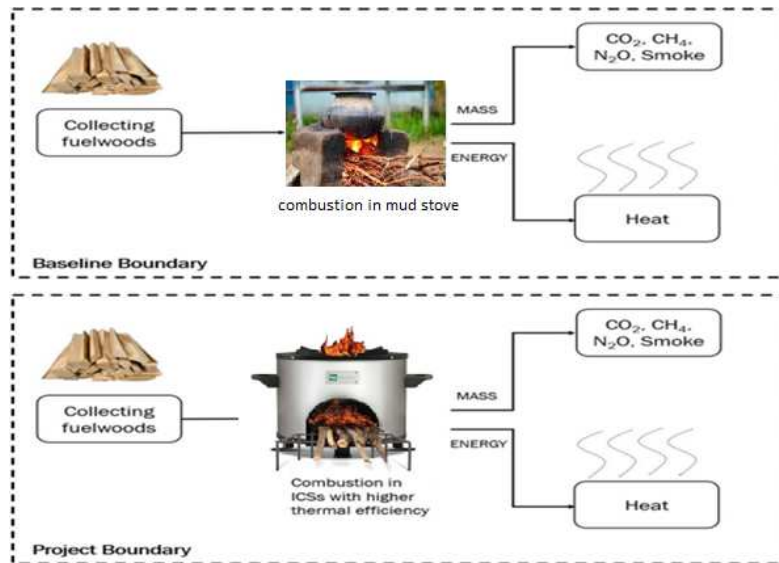


Figure 6 Project Boundary

3.4 Baseline Scenario

The baseline scenario is the continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

As per the National Family Health Survey - 5 (2019-2020) for Meghalaya⁴, around 78.3% households in rural areas are still using non-clean fuel mainly firewood for their daily thermal needs for cooking.

Use of traditional mud cookstoves for cooking and boiling water by the project stove users is deemed as the realistic pre-project scenario. Traditional biomass continues to be the primary energy source for cooking in rural areas and impoverished urban clusters. Approximately two-thirds of households rely on solid cooking fuels, including firewood, crop residue, cow dung cake, and coal/lignite/charcoal⁵. According to the baseline survey conducted by the Project Participant, inefficient cooking practices are widespread in rural areas of Meghalaya.

The survey revealed that households using fuelwood as a fuel for the cooking purposes. It's crucial to note that household air pollution is closely linked to a heightened risk of various health

⁴ [Meghalaya.pdf \(rchiips.org\)](https://rchiips.org/Meghalaya.pdf)

⁵ Kumar, P., & Igdalsky, L. (2019). Sustained uptake of clean cooking practices in poor communities: Role of social networks. *Energy research & social science*, 48, 189-193.

issues, including tuberculosis, asthma, mortality from cardio-respiratory illnesses, and non-respiratory problems such as adverse pregnancy outcomes, prematurity, and low birth weight⁶.

At the national level, statistics provided by the National Institution for Transforming India (NITI Aayog) offer an overview of the baseline energy scenario in India. These statistics underscore the pressing need for interventions and improvements in cooking technologies and energy sources, especially in rural areas, to mitigate the adverse health effects associated with traditional biomass use.

The proportion of population using solid fuels in India was 55.5% (54.8–56.2) in 2017 and 1.24 million (1.09–1.39) deaths in India in 2017, which were 12.5% of the total deaths, were attributable to air pollution⁷. The Project activity are premised on generating Emission Reductions by ensuring that users have cleaner cookstoves, thereby reducing the need for them to burn non-renewable biomass in order to cook food.

There is currently no law or policy which requires the use of fuel-efficient stoves in India. It follows that the Project activity is a voluntary action.

As per the baseline survey conducted in the region, the key reason for current cooking practice of traditional 3-stone chulhas are: easy availability of wood from forest, lack of awareness about the fuel wastage and other harmful effects of low efficiency stoves and the capital cost of the efficient technologies.

The baseline survey was conducted of 120 households from the dates 05-September-2022 to 27-September-2022, and a team of 6 people from the PP was involved in conducting the survey.

3.5 Additionality

The methodology uses the activity method for the demonstration of additionality

Step 1: Regulatory Surplus

There is no mandated government programme or policy in the host country of this project ensuring the distribution of domestic fuel-efficient greenway clean cookstove. The project is not mandated by any law, statute, or another regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute, or other regulatory frameworks. Households may only participate voluntarily in this project. It is hereby confirmed that the proposed project is a voluntary coordinated action by GHE Impact Ventures Private Limited.

Step 2: Positive List

⁶ Jindal, S. K., Aggarwal, A. N., & Jindal, A. (2020). Household air pollution in India and respiratory diseases: current status and future directions. *Current Opinion in Pulmonary Medicine*, 26(2), 128-134

⁷ Balakrishnan, K., Dey, S., Gupta, T., Dhaliwal, R. S., Brauer, M., Cohen, A. J., ... & Sabde, Y. (2019). The impact of air pollution on deaths, disease burden, and life expectancy across the states of India: the Global Burden of Disease Study 2017. *The Lancet Planetary Health*, 3(1), e26-e39

As per methodology, the project must demonstrate that it meets all of the applicability conditions of the methodology as well as the below condition.

1. Where the project activity installs or distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits, the project activity shall be deemed additional.

The project proponent had distributed the greenway Jumbo cookstove at high subsidized cost, and would only charge a sum of INR 200. Hence there will be a revenue stream apart from the sale of GHG credits. Therefore, the project proponent chooses a step 3 for proving additionality.⁸

Note: This project is implemented by the PP using their own funds, and no government support or multilateral funds are used for the implementation of the project.

Step 3: Project Method

In line with Methodology VMR0006: Version 1.1, para 7 step 3 “For any project activity where stoves are not provided at zero cost to the end-user or has any other source of revenues other than the sale of GHG credits, then the project activity shall apply investment analysis method set out in the CDM Tool for the Demonstration and Assessment of Additionality included in AMS-II.G to determine that the proposed project activity is either: 1) not the most economically or financially attractive, or 2) not economically or financially feasible”

Hence, in line with AMS-II.G. Version 13.0 section 5.2; additionality is demonstrated using Option 2, by applying the “TOOL21: Demonstration of additionality of small-scale project activities”; section 5, i.e. *Investment barrier: a financially more viable alternative to the project activity would have led to higher emissions.*

PP has identified realistic and credible alternative scenario(s) to the project activity. The alternative identified are

- (a) The proposed project activity (i.e. distribution of improved cookstove that are efficient and have lower emissions) undertaken without being registered as a VCS project activity;
- (b) Continuation of the current situation (i.e. continuation use of traditional mud stoves which are free of cost but have higher emissions).

Investment Barrier:

Alternative (a) i.e. proposed project activity:

The proposed project intends to distribute the ICS at 94% subsidy to users i.e. households; hence financial return from the programme will be less than cost incurred.

⁸ VMR0006: Methodology for installation of High Efficiency firewood cookstove Version 1.1; page No 8

A sum of 200 rupees is collected from each of the beneficiary for the purpose of a partial contribution towards the cost of cookstove and to ensure effective implementation of project. There is no other revenue generated in the project by the project proponent other than the sale of GHG credits.

Moreover, the recovered cost is clearly documented in the user agreement between PP and cookstove beneficiaries:

S. No	State	Number of cookstoves	Cost recovered from individual beneficiaries	Source of document
01	Meghalaya	20,000	Rs. 200/-	User Agreement

Further each ICS has certain costs associated with the production/purchase, distribution, repairs and maintenance, project management, monitoring and verification etc., the programme will not occur in absence of the carbon revenue.

Hence a **simple cost analysis** is chosen by PP.

PP has carried out simple cost analysis for the demonstration of additionality and from above explanation it is very clear that the ICS has been distributed at a 94% subsidised cost to the end-user or beneficiaries. Hence the proposed project activity is not economically or financially feasible without the revenue from sale of GHG credits.

The below table shows the revenue and subsidy for the Improved cookstove.

S. No	Particular	Cost (INR)t	Source
01	Cost of Improved Cookstove	3,499	Price available on the website ⁹
03	Cost to beneficiary	200	User agreement
04	Percentage subsidy to beneficiary	94%	Calculated

The action is not financially viable without the support of revenues from the sale of VCU. As the end-user does not benefit from a direct financial return and procuring ICSs requires capital, which is a barrier to rural consumers due to difficulties in accessing capital, a wide dissemination of ICSs in the Host Country is unlikely. The actions under the project will alleviate these barriers by promoting distribution of ICSs to end-users at an affordable price.

Alternative (b) i.e. Continuation of the current situation:

Continuation use of traditional mud stoves which are free of cost and financially more viable alternative which leads to higher emissions.

⁹ Cost of Greenway Jumbo Stove <https://www.greenwayappliances.com/shop>

Conclusion:

From the above explanation, it is very clear that the end users have a financially more viable alternative to the project activity i.e. continuation use of traditional mud stoves which are free of cost but have higher emissions. Hence it can be concluded that the project faces Investment Barrier.

3.6 Methodology Deviations

The project did not apply any methodology deviations.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

Under this project activity, the distribution of improved cookstoves has been started from 19-October-2022. Till the end of first monitoring period i.e. on 31-April-2023, 5,000 improved cookstoves has been distributed by the project proponent for Garo hills community.

A total of 20,000 improved cookstoves is estimated to be distributed under this project activity.

5 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

According to the Methodology VMR0006, v 1.1, “Methodology AMS-II.G. does not account for baseline emissions separately, but instead quantifies emission reductions as a function of the reduction in the amount of non-renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves. This revision follows the same convention.” Thus, kindly refer to Section 5.4 for the details of net emission reductions achieved by the project.

5.2 Project Emissions

According to the Methodology VMR0006, v 1.1, “Methodology AMS-II.G. does not account for project emissions separately, but instead quantifies emission reductions as a function of the reduction in the amount of non-renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves. This revision follows the same convention.” Thus, kindly refer to Section 4.4 for the details of net emission reductions achieved by the project

5.3 Leakage

According to the Methodology VMR0006, v 1.1, leakage related to the non-renewable woody biomass saved by the project activity shall be considered as default 0.95

5.4 Estimated Net GHG Emission Reductions and Removals

According to the Methodology VMR0006, v 1.1, net GHG emission reductions are calculated by applying Equation 1 and 2 for project activities replacing baseline stoves using firewood as below.

$$ER_y = \sum_i \sum_j ER_{y,i,j}$$

Equation (1)

Where:

i = indices for the situation where more than one type of cookstove is introduced to replace mud stove fire

j = indices for the situation where there is more than one batch of the improved cookstove of type i

ER_y = Emission reductions during year y in tCO₂e

$ER_{y,i,j}$ = Emission reductions by improved cookstove of type i and batch j during year y in tCO₂e

$$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \times NCV_{wood\ fuel} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times N_{y,i,j} \times 0.95$$

Equation (2)

Where:

$B_{y,savings,i,j}$ = Quantity of woody biomass that is saved in tons per improved cook stove of type i and batch j during year y

$f_{NRB,y}$ = Fraction of woody biomass that can be established as non-renewable biomass ($f_{NRB,y}$)

$NCV_{wood\ fuel}$ = Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/ton)

EF_{wf,CO_2} = CO₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO₂/TJ)

$EF_{wf,non\ CO_2}$ = Non-CO₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO₂/TJ)

$N_{y,i,j}$ = Number of improved cook stoves of type i and batch j operating during year y

0.95 = Discount factor to account for leakage

The quantify of woody biomass saved $B_{y,savings,i,j}$ due to implementation of improved cookstoves can be estimated by Equation 3 below:

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,i,j}}{\eta_{old}} - 1 \right)$$

Equation (3)

Where:

η_{old} = Efficiency of baseline cookstove

$\eta_{new,i,j}$ = Efficiency of the improved cookstove type i and batch j determined through water boiling test (WBT). Alternatively, efficiency may be determined using Equation 4 below

$$\eta_{new,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94$$

Equation (4)

Where:

η_p = Efficiency of project stove (fraction) at the start of project activity.

$(DF_n)_{y-1}$ = Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). This value may be based on actual monitoring or based on

manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards. Alternatively default value of 0.99 efficiency loss per year can be considered.

0.94 = Adjustment factor to account for uncertainty related to project cookstove efficiency test.

f_{NRB} is calculated using the Tool 30: Calculation of the fraction of non-renewable biomass Version 04.0

the fraction of woody biomass that can be established as non-renewable is

$$f_{NRB} = \frac{NRB}{NRB + RB}$$

Where,

NRB = Quantity of non-renewable biomass consumed in the applicable areas in relevant period (tonnes)

RB = quantity of renewable biomass that is available on a sustainable basis in the applicable area in the relevant period (tonnes)

$$NRB = H - RB$$

Where

H = Total consumption of woody biomass in the applicable area in the relevant period. (tonnes)

$$H = HW \times N + CE + NE$$

Where;

HW	=	Average consumption of wood fuel per household, including fuelwood and charcoal, in the applicable area in the relevant period (tonnes//household)
CE	=	Commercial woody biomass consumption for energy applications (e.g. commercial, industrial or institutional uses of woody biomass in ovens, boilers etc.) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)
NE	=	Commercial woody biomass consumption for non-energy applications (e.g. construction, furniture) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)
N	=	Number of households consuming wood fuel within the applicable area in the relevant period (number)

$$RB = \sum (MAI_{forest,i} \times (F_{forest,i} - P_{forest,i})) + \sum (MAI_{other,i} \times (F_{other,i} - P_{other,i}))$$

$MAI_{forest,i}$	=	Mean Annual Increment of woody biomass growth per hectare in sub-category i of forest areas in the relevant period (tonnes/ha/yr)
$MAI_{other,i}$	=	Mean Annual Increment of woody biomass growth per hectare in sub-category i of other land areas in the relevant period (tonnes/ha/yr)
$F_{forest,i}$	=	Extent of forest in sub-category i in the relevant period (ha)
$F_{other,i}$	=	Extent of other land in sub-category i in the relevant period (ha)
$P_{forest,i}$	=	Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within forest areas (in sub-category i) in the relevant period (ha)
$P_{other,i}$	=	Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within other land areas (in sub-category i) in the relevant period (ha)
i	=	Sub-category i of forest areas and other land areas

$$RB = 0.740(1,713,718 - 294,600) + 2.267(113,390 - 0)$$

$$RB = 1,851,871.149 \text{ tonnes}$$

$$H = (2.4 \times 2,154,880) + 0 + 4,392.6$$

$$H = 5176104.6 \text{ tonnes}$$

H	Calculated	5,176,104.6
RB	Calculated	1,851,871.149
NRB	Calculated	3,324,233.451
FNRB		64.22%

The significant difference between the values of f_{NRB} achieved by the Bailis report the carbon footprint of traditional woodfuels (25.8%) and the values achieved by our project (64.22%). To address the feasibility of the value calculated based on CDM Tool 30 version 4 and its alignment with the conservative principle in the VCS Standard section 2.2, the following points should be considered:

1. Data Source and Accuracy:

VCS 4250: The project's f_{NRB} calculation relies on the most recent data of forest cover, fuelwood consumption, annual increment from the “Forest Survey of India” report published by Ministry of Environment Forest and Climate Change (MoEFCC). The report is published every two years and

the latest report referred in the project is of 2021 (FSI)¹⁰. While the demographic details (family size, household data using fuel wood for cooking) are based on the National Family Health Survey (NFHS) report, NFHS surveys have been conducted under the stewardship of the Ministry of Health and Family Welfare (MoHFW), Government of India. MoHFW has designated the International Institute for Population Sciences (IIPS) which publishes the survey report. The project applies the data from the latest available report of NFHS-5¹¹ 2021. The f_{NRB} value is calculated based on the above-mentioned reports specifically for the region “State of Meghalaya”, i.e., project boundary.

Bailis report: The report uses global-level data from the Food and Agriculture Organization (FAO), the International Energy Agency (IEA) and the United Nations (UN)¹² for various data along with other published literature for specific countries.

- 1) For specific f_{NRB} data for India, Bailis report estimates the NRB values of the region using the WISDOM Model¹³ which takes the values of supply and demand module of a given spatial base of the region, and reliance on the local surveys for the same was one of the parameters in the model. Since, there was no survey conducted in the region for the demand of the woodfuel, the Bailis report has referred to the published data of woodfuel consumption for India as per two reference literature;
 - a) Paper titled “Comparative study of fuelwood consumption by villagers and seasonal “Dhaba owners” in the tourist affected regions of Garhwal Himalaya, India”. This report refers to wood fuel consumption for “Dhaba owners” i.e., small restaurants in local language. The region of study was tourist affected region of Garhwal which is in Uttarakhand state of India. This paper was published in year 2010.
 - b) Paper titled “Firewood consumption along an altitudinal gradient in mountain villages of India”. This report refers to fuel wood consumption of village households in Uttaranchal state of India. This paper was published in year 2004.

Inference: The data used by Bailis report specifically for India is based on a paper of 2004 and 2010, thus not representing the latest data as published by the national agencies as well as

¹⁰ The data of the Forest Survey of India is published regionally as well as at the national level. The latest reports for India and the project boundary could be downloaded from <https://fsi.nic.in/fsiwebsite/forest-report-2021>

¹¹ National Family Health Survey is conducted every two years and the latest published data are provided on the link http://rchiips.org/nfhs/NFHS-5_State_Report.shtml. This report was published in the year 2021.

¹² As mentioned in the section ‘Methods’ of the paper: The carbon footprint of Traditional Woodfuels.

¹³ Woodfuel Integrated Supply/Demand Overview Mapping (WISDOM) Model which estimates the value of the fuelwood consumptions in the áreas. <https://www.sciencedirect.com/science/article/abs/pii/S0961953406000262>

regional data available for the project boundary, i.e., State of Meghalaya. There is considerable difference in the values as published by the national agencies. Further Bailis report does not take into account the values for the project boundary, i.e. State of Meghalaya.

2. **Adherence to Tool 30:** The project applies VMR0006 version 1.1 which refers the calculation of f_{NRB} based on latest applicable version of Tool 30 as published by CDM EB, i.e version 4.0. The tool provides detailed steps to calculate the f_{NRB} value for a given region.

VCS 4250: The project has detailed step wise calculation for assessment of f_{NRB} values. The value has been calculated for the applicable project boundary which is limited to the state of Meghalaya by using latest regional level data as published by the various government agencies. The f_{NRB} value arrived is 64.22% for the project boundary.

Bailis Report: The report is based on WISDOM Model. The f_{NRB} values published for India is 23-25%. The tool also published a default value of f_{NRB} value that can be used which is 30%.

Inference: the f_{NRB} value as calculated by the VCS 4250 is in adherence to the Tool 30 version 4.0.

3. Limitations of Bailis Report : As published in the Bailis report on Carbon footprint of traditional woodfuels page number 4 we would like to highlight the below:

Section "Discussion and Implementation" mentions ***"One limitation of the study is a lack of reliable woodfuel consumption data. When possible, we used national and sub national data sets. However, for most countries, we relied on data compiled by international organizations containing unknown uncertainties that make it difficult to communicate the uncertainty in these results. A second limitation is that the analysis considers a single year and does not account for potential behavioural changes among woodfuel users"***.

Inference: The Bailis report is not a correct representative of the f_{NRB} for the given project boundary, i.e. State of Meghalaya in India.

4. Values published in other projects: There are various projects with similar project boundary of state of Meghalaya and also the whole country, a summary of f_{NRB} values in the same is mentioned below:

VCS: details about other VCS projects in the state:

VCS ID	f_{NRB} values	Date of Registration	Project Boundary
2908	74%	30/May/2023	Meghalaya
2533	90.10%	02/July/2022	India

2473	93.10%	09/June/2022	India
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Inference: The f_{NRB} values are comparable to other similar projects in the region, which has been registered by the VERRA and Gold Standard Mechanism and VVB. Therefore, the conservative principle does not apply in this context.

Conclusion: There are considerable differences in the approach and value determined by the project as compared to the Bailis report. Since the value of f_{NRB} as calculated in the project is based on latest specific regional area of country, i.e. State of Meghalaya thus it represents actual scenario as compared to f_{NRB} as reported by Bailis report which is for the country. The report itself claims data to be from 2004 and 2010 thus it does not represent the actual scenario. The report fails to mention the uncertainty in the results. Further, the value is calculated for a single year and does not take into account the trend in the past decade.

We would thus like to mention that the f_{NRB} value as calculated in the project is appropriate for the given project boundary, i.e. State of Meghalaya.

Conservative assessment: The comparison with Bailis report is not applicable since the data and uncertainty for f_{NRB} values in Bailis report are neither published nor reliable for the project boundary, i.e. State of Meghalaya.

The parameters and calculations of emission reductions with reference to above parameters is given below as

Using equation (4)

$$\begin{aligned}\eta_{new,i,j} &= \eta_p \times (DF_n)^{y-1} \times 0.94 \\ \eta_{new,i,j} &= 38\% \times (0.99)^{1-1} \times 0.94 \\ \eta_{new,i,j} &= 35.72\%\end{aligned}$$

Using Equation (3)

$$\begin{aligned}B_{y,savings,i,j} &= B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,i,j}}{\eta_{old}} - 1 \right) \\ B_{y,savings,i,j} &= 1.46 \times \left(\frac{35.72}{10} - 1 \right) \\ B_{y,savings,i,j} &= 3.76 \text{ tonne/ICS} - \text{year}\end{aligned}$$

Parameters	Value	Unit	Source
Wood Used by ICS	4	kg/day	Sample Survey
$B_{Y=I_{new, I, J, survey}}$	1.46	tonne/ICS/Year	Monitoring Survey
Total Days in a year	365	days	
η_{old}	10%	(For mud stove)	Default Value as per Methodology VMR0006 V1.1 -Page no. 19
η_p	38%	%	Manufacturer's Specification
$B_{Y savings I, J}$	3.76	tonne/ICS-year	Calculated
f_{NRB}	64.22%	%	f_{NRB} calculation sheet as per the Tool 30
$NCV_{wood fuel}$	0.0156	TJ/tonne	Default Value as per Methodology VMR0006 V1.1 -Page no. 23
EF_{wf, CO_2}	112	tCO ₂ /TJ	Default Value as per Methodology VMR0006 V1.1 -Page no. 10
$EF_{wf, nonCO_2}$	26.23	tCO ₂ /TJ	Default Value as per Methodology VMR0006 V1.1 -Page no. 10
$N_{y, I, J}$	20,000	Number	Number of cookstoves
Discount Factor	0.95	fraction	To account for leakage, Default Value as per Methodology VMR0006 V1.1 -Page no. 09
$(DF_n)^{y-1}$	0.99		Default Value as per Methodology VMR0006 V1.1 -Page no. 11
Adjustment Factor	0.94		Adjustment factor to account for uncertainty related to project cookstove efficiency test

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
19-October-2022 to 31-December-2022	0	20,032	0	20,032
01-January-2023 to 31-December-2023	0	97,436	0	97,436
01-January-2024 to 31-December-2024	0	96,077	0	96,077
01-January-2025 to 31-December-2025	0	94,732	0	94,732
01-January-2026 to 31-December-2026	0	93,401	0	93,401

01-January-2027 to 31-December-2027	0	92,083	0	92,083
01-January-2028 to 31-December-2028	0	90,778	0	90,778
01-January-2029 to 18-October-2029	0	78,776	0	78,776
Total		663,315		663,315

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	f_{NRB}
Data unit	%
Description	Fraction of woody biomass saved by the project activity during year y that can be established as non-renewable biomass.
Source of data	Determined by using Tool 30: Calculation of the fraction of non-renewable biomass. Version:04.0
Value applied:	64.22%
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante
Purpose of Data	Calculation of baseline emission reduction
Comments	NA

Data / Parameter	$NCV_{wood\ fuel}$
Data unit	TJ/tonne
Description	The net calorific value of the non-renewable woody biomass that is substituted or reduced
Source of data	Default as per applied approved methodology VMR0006 ver 1.1 Page number 11
Value applied:	0.0156
Justification of choice of data or description of measurement methods and procedures applied	The Default value as per methodology VMR0006, ver1.1, page number 11

Purpose of data	Calculation of emission reductions
Comments	This parameter is fixed ex-ante

Data / Parameter	η_{old}
Data unit	%
Description	The efficiency of baseline cookstove
Source of data	default value
Value applied:	10%
Justification of choice of data or description of measurement methods and procedures applied	A default value of 0.1 shall be used if baseline device is a mud stove fire using fuelwood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney.
Purpose of data	<i>Calculation of emission reductions</i>
Comments	This value is fixed ex-ante

Data / Parameter	η_p
Data unit	%
Description	The efficiency of the project stove at the start of project activity
Source of data	Manufacturer's Specification
Value applied:	38%
Justification of choice of data or description of measurement methods and procedures applied	Manufacturing Specification as per the 3 rd party WBT test.
Purpose of Data	Calculation of η_{new}

Comments	This parameter is fixed ex-ante for the Greenway Jumbo cookstove.																
Data / Parameter	$\eta_{new,y,i,j}$																
Data unit	Fraction																
Description	Efficiency of the improved cookstove type i and batch j determined through water boiling test (WBT) during year y																
Source of data	Calculation																
Description	To adopt Option V given in the methodology: “Efficiency of the improved cookstoves to be estimated using equation 5 above where loss in efficiency per year is calculated, and therefore this parameter does not need to be monitored”																
Value applied	For ex-ante calculation, the value below is applied <table border="1"> <thead> <tr> <th>Year(y)</th><th>$\eta_{new,y,i,j}$</th></tr> </thead> <tbody> <tr><td>1</td><td>35.72%</td></tr> <tr><td>2</td><td>35.36%</td></tr> <tr><td>3</td><td>35.01%</td></tr> <tr><td>4</td><td>34.66%</td></tr> <tr><td>5</td><td>34.31%</td></tr> <tr><td>6</td><td>33.97%</td></tr> <tr><td>7</td><td>33.63%</td></tr> </tbody> </table>	Year(y)	$\eta_{new,y,i,j}$	1	35.72%	2	35.36%	3	35.01%	4	34.66%	5	34.31%	6	33.97%	7	33.63%
Year(y)	$\eta_{new,y,i,j}$																
1	35.72%																
2	35.36%																
3	35.01%																
4	34.66%																
5	34.31%																
6	33.97%																
7	33.63%																
Justification of choice of data or description of measurement methods and procedures applied	Calculated value from Equation 5 in VMR0006 ver.1.1 (No Monitoring equipment used)																
Purpose of data	Calculation of emission reductions																
Comments	No Comments																

Data / Parameter	$EF_{wf,CO2}$
Data unit	tCO ₂ /TJ

Description	CO ₂ emission factor for the use of wood fuel in baseline scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion ¹⁴
Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	<i>Calculation of emission reduction</i>
Comments	NA

Data / Parameter	EF _{wf, non-CO2}
Data unit	tCO ₂ /TJ
Description	Non-CO ₂ emission factor for the use of wood fuel in baseline scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion
Value applied	26.23
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of emission reduction
Comments	NA

¹⁴ https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_2_Ch2_Stationary_Combustion.pdf

6.2 Data and Parameters Monitored

Data / Parameter	$N_{y,i,j}$
Data unit	Number
Description	Number of project devices of type i and batch j operating during year y
Source of data	Monitoring
Description of measurement methods and procedures applied	Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 90/10 confidence precision according to the latest version of Standard for sampling and surveys for CDM project activities and program of activities.
Frequency of monitoring/recording	At least once every two years
Value applied:	20,000
Monitoring equipment	Monitoring survey
QA/QC procedures applied	Sampling standard "sampling and surveys for CDM project activities and program of activities" version 9 will be used for determining the sample size to achieve 90/10 confidence precision.
Purpose of data	Calculation of emission reductions.
Calculation method	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value
Comments	No Comments

Data / Parameter	$B_{y=1.new.i,j.survey}$
Data unit	Tonnes/device/year
Description	Annual quantity of woody biomass used by improved cookstoves in tonnes per device of type i and batch j, determined in the first year of the implementation of the project through a sample survey
Source of data	Monitoring survey
Description of measurement methods	Minimum sample size of each type i and batch j should be in line with the latest version of Standard for sampling and surveys for

and procedures to be applied	CDM project activities and program of activities or guidelines provided in methodology Section 8.4 option (b). Determined in the first year of the introduction of the devices (e.g., during the first year of the crediting period, $y=1$) through measurement campaigns at representative households and/or sample survey. Sample surveys to estimate this parameter, that are solely based on questionnaires or interviews (i.e., that do not implement measurement campaigns) may only be used if the following conditions are satisfied. (i) Baseline cookstoves have been completely decommissioned and only improved cookstoves are exclusively used in the project households; The determination of the parameter was based on the measurement campaigns, where the sampled beneficiaries were issued amount of the wood biomass in the morning and the left wood was weighed at the end of day. This is how the consumption of the biomass was determined by the project proponent.
Frequency of monitoring/recording	Determined in the first year of project implementation during the first monitoring period that is from 19/October/2022 to 30/April/2023.
Value applied	1.46 tonnes/device/year.
Monitoring equipment	Weighing Scales to be used for measuring quantity of fuel wood. 4 weighing scales were used. Their serial numbers are SR1042356, SR1041429, SR1046547, SR1042759.
QA/QC procedures to be applied	New set of weighing scale would be purchased for each monitoring.
Purpose of data	Calculation of emission reductions
Calculation method	Not Applicable. This is monitored value. The sampled beneficiaries were issued a quantity of woodfuel in the morning and the leftover wood was weighed at the end of the day. This is how the project proponent determined the consumption of woodfuel.
Comments	The monitoring of this parameter happened on the 01-May-2023 to 09-May-2023
Data / Parameter	Life Span

Data unit	Number of years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the project stove efficiency
Source of data	Manufacturer's specification
Description of measurement methods and procedures to be applied	Technical specification as declared by the manufacturer and wherever possible supported by third party report.
Frequency of monitoring/recording	Once at the time of project stove installation
Value applied	7 Years
Monitoring equipment	No equipment is required to monitor this parameter
QA/QC procedures to be applied	This parameter will be referred from the Manufacturer's specification
Purpose of data	Calculation of emission reductions
Calculation method	No calculation used for this parameter
Comments	If the life span of project devices is less than the crediting period, it shall be demonstrated that the devices shall be replaced after the life span has ended. In such cases, if it cannot be demonstrated that the project devices will be replaced with new devices, no emission reductions can be claimed beyond the life span of the project devices.

6.3 Monitoring Plan

Sampling Plan

As per the Guideline for Sampling and Surveys for CDM Project Activities and Program of Activities, version 04.0, the sampling plan is the following:

(a) Sampling design

Due to the large number of Improved Cookstove distributed under the project, it is not economically feasible to monitor each individual Improved Cookstove unit distributed. Therefore, representative sampling shall be undertaken for each batch of Improved Cookstove. The

sampling plan to be designed in line with the requirements of the Standard for Sampling and Surveys for CDM Project Activities and Program of Activities, version 04.0. Samples will be drawn from ICS recorded in the project database administered by the project proponent. A detailed explanation of this project is described in Section 1.11 of this document.

(i) **Objectives and Reliability Requirements**

The objective is to obtain an unbiased and reliable estimate of the proportion or mean value of the following key variables over the course of the crediting period. As per CDM Methodology AMS-II.G, 90/10 confidence/precision shall be applied for annual sampling requirement and 95/10 for biennial sampling inspection. As per Standard for Sampling and Surveys for CDM Project Activities and Program of Activities, 90/10 confidence/precision to be adopted for small-scale project and 95/10 for large-scale project. Given that the size of the project is under the category of projects, hence 90/10 confidence/precision shall be adopted for all parameters unless the average annual emission reductions of the project below the threshold.

Monitored Parameters:

The project involves introduction of improved cook stoves as energy efficiency measure:

No.	Monitoring Parameters	Sampling Parameters	Parameter type	Monitoring Frequency
1.	$N_{y,i,j}$ Number of project devices of type i and batch j operating during year y	Proportion of ICS still in operation	Proportion	Biennially
2.	$B_{y=1.new.i.j.survey}$ Quantity of woody biomass used by project devices in tonnes per device of type i. and batch j	Daily consumption of woody biomass per ICS	Mean value	Determined in the first year of project implementation

(ii) **Target Population**

The target population will be the complete set of appliances (ICS) deployed under the project.

(iii) **Sampling Method**

The project involves distribution of Improved Cookstove throughout the project area thereby replacing traditional cooking devices. The population is homogeneous in nature i.e., common technology with similar operating characteristics and similar food habits. Therefore, Simple Random Sampling technique will be used to conduct sampling survey among Improved Cookstove batches.

The populations of each batch will be combined together, the sample size is determined, and a single survey will be undertaken to collect data. To ensure the survey result is representative of the entire population, the dissimilarity (such as Improved Cookstove type, vintage and location in which they are operating) within the included project will be taken into account in the sample size calculation. The Improved Cookstove of same type, vintage and location in which they are operating will be grouped in the same batch. Samples will be drawn by using the random number generator.

To determine the parameters, sampling will involve the following approaches (outcome in brackets):

$N_{y,t,j}$ = Visual inspection of the premises to see if Improved Cookstove is operational and in use. Interview with end user if required to verify that Improved Cookstove is still in use (Yes/No)

$B_{y=1,new,t,j,survey}$ = Interview with end user and estimate the daily consumption of woody biomass of Improved Cookstove (Daily consumption of woody biomass)

Sample Size

The procedure to determine the sample of households will ensure that they adequately represent the broader project population, minimizing sampling error. Using, a 90 per cent confidence level, and a 10 per cent margin of error, a random sample will be selected. In order to calculate the required sample size estimates, values for the proportions, mean values, and standard deviations are required. As per Guideline for Sampling and surveys for CDM project activities and programmers of activities, version 04.0, Sample size calculations for small scale; Simple random sampling.

$$n \geq \frac{1.645^2 N \times p(1-p)}{(N-1) \times 0.1^2 \times p^2 + 1.645^2 p(1-p)}$$

Where:

n = Sample Size

N = Total Number of households (5,000)

p = Our expected proportion (0.80)

1.645 = Represents the 90% confidence interval

0.1 = Represents 10% relative precisions ($0.1 \times 0.5 = 0.05 = 5\%$ points on either side of p)

(b) Data to be collected:

(i) Field Measurements:

The table below summarizes field measurement data requirements

Parameter	Parameter	Frequency as required by methodology	Methods to be applied	Comments on seasonal fluctuation

$N_{y,i,j}$	Monitoring will likely occur every 12 to 24 months	Biennially	Visits to the premises, visual inspection and interview with ICS end-user.	Not due to any seasonal fluctuation.
$B_{y=1,new,i,j,survey}$	Monitoring will likely occur within the first year after installation	First year of installation	Visits to the premises, visual inspection and interview with ICS end-user.	Not due to any seasonal fluctuation.

Surveys will be conducted and data collection would be done by using the mobile applications, all the data collected would be kept and the database will have back up. Original stove distribution data, declaration data, information collected from the registration or other means of acceptance by the users will be stored in the project office. A back-up of the project database will also be stored on an electric medium. All data monitored and required for verification and issuance will be kept for two years after the end of the crediting period or the last issuance of credits for the project activity, whichever is later.

(ii) Quality Assurance/Quality Control

The project proponent will apply measures to ensure the required confidence/precision for each sampled parameter is met, allowing for non-response and the possible removal of outliers from the sample, as part of a Quality Control/Quality Assurance system. The choice of measure applied to each parameter will depend on the cost of each data collection approach and logistics required. The project proponent will determine the most effective measure for each parameter from the following list:

- Oversampling: Randomly draw a sample more than the calculated number (say 20%) and collect data from each
- Buffer Group: Randomly draw additional samples (say 20%) and collect data from only for minimum numbers of ICS as per sample size calculation. If this would not result in the required sample size data would be collected from the additional ICS that were selected in the sample.
- If precision required is not achieved by reliability check, additional data collection that is a supplemental or new sample would be taken to reach the required precision level
- A quality control and assurance strategy will be documented. Quality control and assurance strategies include addressing non-sampling errors, such as non-response or bias from interviewer. The project proponent or a competent third party designated by the project proponent with the proper skills will train the monitoring personnel on how to properly survey households to prevent bias from interviewer. In the case a household refuses to participate, another household will be chosen at random. To reduce interviewer bias, good questionnaire design and well-tested questionnaires will be used. The sample data for mean value

parameter is continuous and therefore the presence of outliers is possible. To identify and address outliers for the parameter, outliers will be defined as those data points with values greater than three standard deviations from the mean of the sample. Data points identified as outliers will be examined further to correct for possible transcription and data entry errors but will be omitted from the analysis if no such administrative errors exist.

(iii) Analysis

The project proponent will manage a project database that includes the following data that can be directly attributable to each batch within the project, thereby allowing unambiguous determination of the emission reductions attributable to each project:

- A list of households participating in each batch/project, including name, community/location, distribution/installation date and unique serial number;
- Testing to ensure that the stoves are still operating above the minimum 25% efficiency required by the methodology, by the project proponent or a third party certified by a national standards body or an appropriate certifying agency recognized by it.
- Where replacements are made, assurance that the efficiency of the new ICS is similar to the specified.

Data obtained from the samples will be used to estimate proportions and mean values for the parameters described above. The values will then be factored into the emissions reduction calculations and result in the request for issuance of VERs. The parameters are applied for emission reduction calculations. The stoves that are not in use will be excluded from emissions reductions calculations and will not be counted towards the total number of ICS in operation during the monitoring period.

(c) Implementation

Sampling for the purpose of emission reduction calculation and elaboration of the monitoring report will occur at the end of each monitoring period.

This sampling will be conducted by trained personal from project proponent or an experienced third-party entity. The credentials and/or training materials for the sampling personal will be provided to the VVB at verification. The maximum length of one monitoring period will be two years (duration, not calendar years), with option for annual or bi-annual monitoring.

The project proponent will be responsible for managing household data collection and entry into the project database. Field personnel will receive training on how to properly deal with surveying techniques and reduce errors and sign a document certifying that there is no conflict of interest of those involved in data collection and analysis. If there is conflict of interest, the personnel will not be allowed to participate in data collection and analysis.

The project database will record the start and end dates of each monitoring period and record the emission reductions attributable to each monitoring period. Appropriate record keeping procedures will be implemented to ensure that each monitoring period data set can be transparently attributed to its corresponding batch/project, preventing any occurrences of double counting. An internal

review of the project database will be able to determine the current status of each batch – the duration of previous monitoring periods, the households delivering monitoring data, and current verification activities. (i) Assessment for Leakage The methodology provides a net to gross adjustment factor of 0.95 to account for leakages, hence the surveys are not required to determine leakage. The other source of leakage occurs if equipment currently being utilized is transferred from outside the boundary to the project activity. All ICS in the project will be newly manufactured/assembled. (ii) Disposal of Low Efficiency Appliances and Use of Baseline Stoves When an ICS is distributed; the end user receives information explaining that conventional open fire appliance must no longer be used. Follow-up meetings with end users will ensure that those who have received an ICS are using it properly and that the conventional open fire is no longer in use. If it is determined that the conventional open fire is still in use along with the operating ICS, a survey will be conducted.

(iv) Monitoring Reporting

The project proponent will assess all monitoring data and produce a monitoring report for the VVB to verify corresponding to the preceding monitoring period. This report will present the data relating to the emission reductions generated by those batches during the monitoring period

For the internal auditing and quality assurance/quality control procedures, the following steps are taken:

1. An internal audit team in GHE Impact Ventures is appointed, which will conduct periodically meetings to ensure that the monitoring plan is being implemented effectively and efficiently.
2. The audit team will review the monitoring plan, data collection procedures, data management, and reporting processes.
3. The team will identify any non-conformances and suggest corrective actions to be taken by the implementing partners.
4. The implementing partners will take corrective actions to address the identified non-conformances.
5. The corrective actions taken will be documented and reported to the oversight agency for review and approval.

The internal audit and quality control procedures will be an ongoing process to ensure that the project is implemented as per the monitoring plan and is meeting its objectives.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

The first monitoring period for the project is dated from 19/October/2022 to 30/April/2023 i.e. total of 194 days of operations.

The monitoring of the parameters was carried out on the sampled cookstoves from dates 01/May/2023 to 09/May/2023.

7.1 Data and Parameters Monitored

Data / Parameter	$N_{y,i,j}$
Data unit	Number
Description	Number of project devices of type i and batch j that are operating in year y
Value applied:	5000
Comments	Improved Cookstoves are distributed in Garo hill districts of Meghalaya

Data / Parameter	$B_{y=1,new,i,j,survey}$
Data unit	Tonne/device/year
Description	Quantity of woody biomass used by project device in tonne per device of type i
Value applied:	0.67
Comments	Survey was carried out with minimum sample size. The measurement campaigns were organized and the consumption of the woody biomass in the project stove was determined. The sampled beneficiaries were issued amount of the wood biomass in the morning and the left wood was weighed at the end of day. This is how the consumption of the biomass was determined by the project proponent.

	There was use of new weighing scales for the purpose of monitoring. The value is for this crediting period which determined in the first year of introduction of device.
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Data / Parameter	Life span
Data unit	Number of Years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the project stove efficiency.
Value applied:	7
Comments	Manufacturer's specification for Greenway jumbo cookstove

7.2 Baseline Emissions

The methodology does not account for baseline emissions separately, but instead quantifies emission reductions as a function of the reduction in the amount of non-renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves.

7.3 Project Emissions

The methodology does not account for project emissions separately, but instead quantifies emission reductions as a function of the reduction in the amount of non-renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves.

7.4 Leakage

Leakage shall be considered as default 0.95 in accordance with Section 5.4 of AMS-II.G.

7.5 Net GHG Emission Reductions and Removals

The improved cookstove is introduced as energy efficiency measure in the project, therefore equations 1 and 2 of the methodology will be applied to calculate the net GHG emission reductions.

$$ER_y = \sum_i \sum_j ER_{y,i,j} \dots\dots\dots \text{Equation 1}$$

Where:

i = Indices for the situation where more than one type/model of improved cookstove is introduced to replace mud stove fire

j = Indices for the situation where there is more than one batch of improved cookstove of type *i*

ER_y = Emission reductions during year *y* in t CO₂e

$ER_{y,i=1,j}$ = Emission reductions by improved cookstove of type *i* and batch *j* during year *y* in tCO₂e And,

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{woodfuel} \times f_{NRB} \times (EF_{wf,CO2} + EF_{wf,nonCO2}) \times N_{y,i,j} \times 0.95$$

Where,

$B_{y,savings,i,j}$ = Quantity of woody biomass that is saved in tonnes per improved cookstove of type *i* and batch *j* during year *y*

$NCV_{woodfuel}$ = Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne)

f_{NRB} = Fraction of woody biomass that can be established as non-renewable biomass f_{NRB} value

$EF_{wf,CO2}$ = CO₂ emission factor for the use of wood fuel in the baseline scenario (IPCC default for wood fuel, 112 tCO₂/TJ)

$EF_{wf,nonCO2}$ = Non-CO₂ emission factor for the use of wood fuel in the baseline scenario (IPCC default for wood fuel, 26.23 tCO₂/TJ)

$N_{y,i,j}$ = Number of improved cookstoves of type *i* and batch *j* operating during year *y*

0.95 = Discount factor to account for leakage

The quantify of woody biomass saved $B_{y,savings,i,j}$ due to implementation of improved cookstoves is estimated by the following Equation:

$$B_{y,saving,i,j} = B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,i,j}}{\eta_{old}} - 1 \right)$$

Equation (3)

Where:

η_{old} = Efficiency of baseline cookstove (10%)

$\eta_{new,i,j}$ = Efficiency of the improved cookstove type *i* and batch *j* determined through water boiling test (WBT) during year *y*, Alternatively, efficiency may be determined using Equation 5 of applied approved methodology (efficiency of Improved cookstove is 38%)

$B_{y=1,new,i,survey}$ = Annual quantity of woody biomass used by improved cookstoves in tonnes per device of type i and batch j , determined in the first year of the implementation of the project through a sample survey.

Example of calculation for woody biomass saving

$$B_{y,saving,i,j} = 0.670 \times \left(\frac{0.357}{0.10} - 1 \right)$$

$$B_{y,saving,i,j} = 1.725 \text{ tonnes}$$

For emission reductions

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{woodfuel} \times f_{NRB} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times N_{y,i,j} \times 0.95$$

$$ER_{y,i,j} = 1.725 \times 0.0156 \times 0.64 \times (112.0 + 26.23) \times 5,000 \times 0.95$$

$$ER_{y,i,j} = 11,376 \text{ tCO}_2e$$

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
19-October-2022 – 31-December-2022	0	1,211	0	1,211
1-January-2023 – 30-April-2023	0	10,165	0	10,165
Total	0	11,376	0	11,376

<u>Ex-ante emissions reductions /removals</u>	<u>Achieved emissions reductions /removals</u>	<u>Percent difference</u>	<u>Justification for the difference</u>

13,129	11,376 ¹⁵	-13%	Distribution of the cookstove was over a period of time. Because of which the actual ER generated by the 5000 cookstoves during this monitoring period is less than that of estimated. More cookstoves would be distributed during the subsequent years.
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¹⁵ The difference in the actual ER is fact due to the calculation based on ex-ante value used in estimation of the ERs. For further information kindly refer to Actual ER sheet.

APPENDIX- 1

Monitoring Survey Form

Global Himalayan Expeditions

Cookstove Auditing Form

Date: _____

1. Volunteer name: _____
2. Cookstove Serial Number: _____
3. Date of Distribution: _____

A. Household Information

1. Name of beneficiary: _____
2. Father/husband name: _____
3. Village: _____
4. Block: _____
5. District: _____
6. Family Members: _____

B. Cooking Details

1. Number of times cooking is done in a day: _____
2. Per day wood consumption in improved cookstove: _____
3. Is their time saving by cooking on improved cookstove: Yes/No
4. Does traditional cookstove is still in use? Yes/No
5. Is the smoke from the improved cookstove is less in amount? Yes/No
6. Is there any concern related to use of improved cookstove? _____

Signature of the beneficiary
